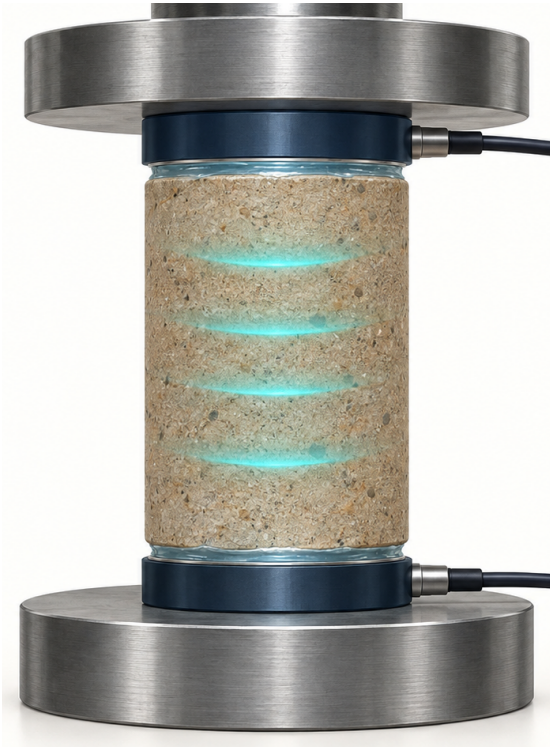
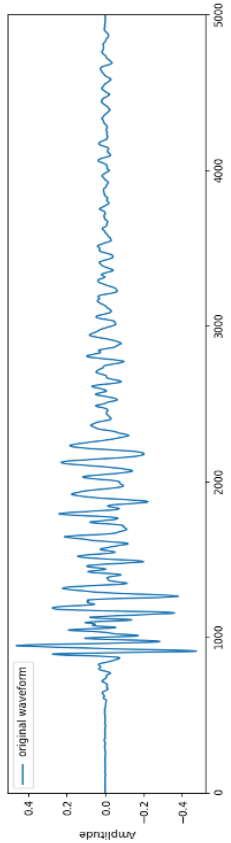
Elastic waves as rock-state probes

From velocity to dynamic moduli, cracks, fluids and damage

Petrophysics · Lecture 5





What changes inside a rock when its waveform changes?

Learning objectives and lecture sequence

By the end of this lecture, students should be able to:

01 derive dynamic elastic moduli from V_p, V_s and density

02 predict how crack closure, crack growth and saturation affect wave observations

03 separate travel-time evidence from amplitude and waveform evidence

04 construct a defensible rock-state interpretation with explicit assumptions

SEQUENCE

dynamic moduli → cracks and stress → fluids → attenuation → integrated diagnosis

Static and dynamic moduli probe different responses

STATIC TEST

Apply a macroscopic load and measure the resulting deformation

- Modulus from the slope of the stress–strain curve
- Finite strain over a selected loading interval
- Captures the bulk response: crack closure, grain-contact deformation, frictional sliding and possible irreversible strain

DYNAMIC TEST

Send a small elastic wave through the rock and measure its velocity

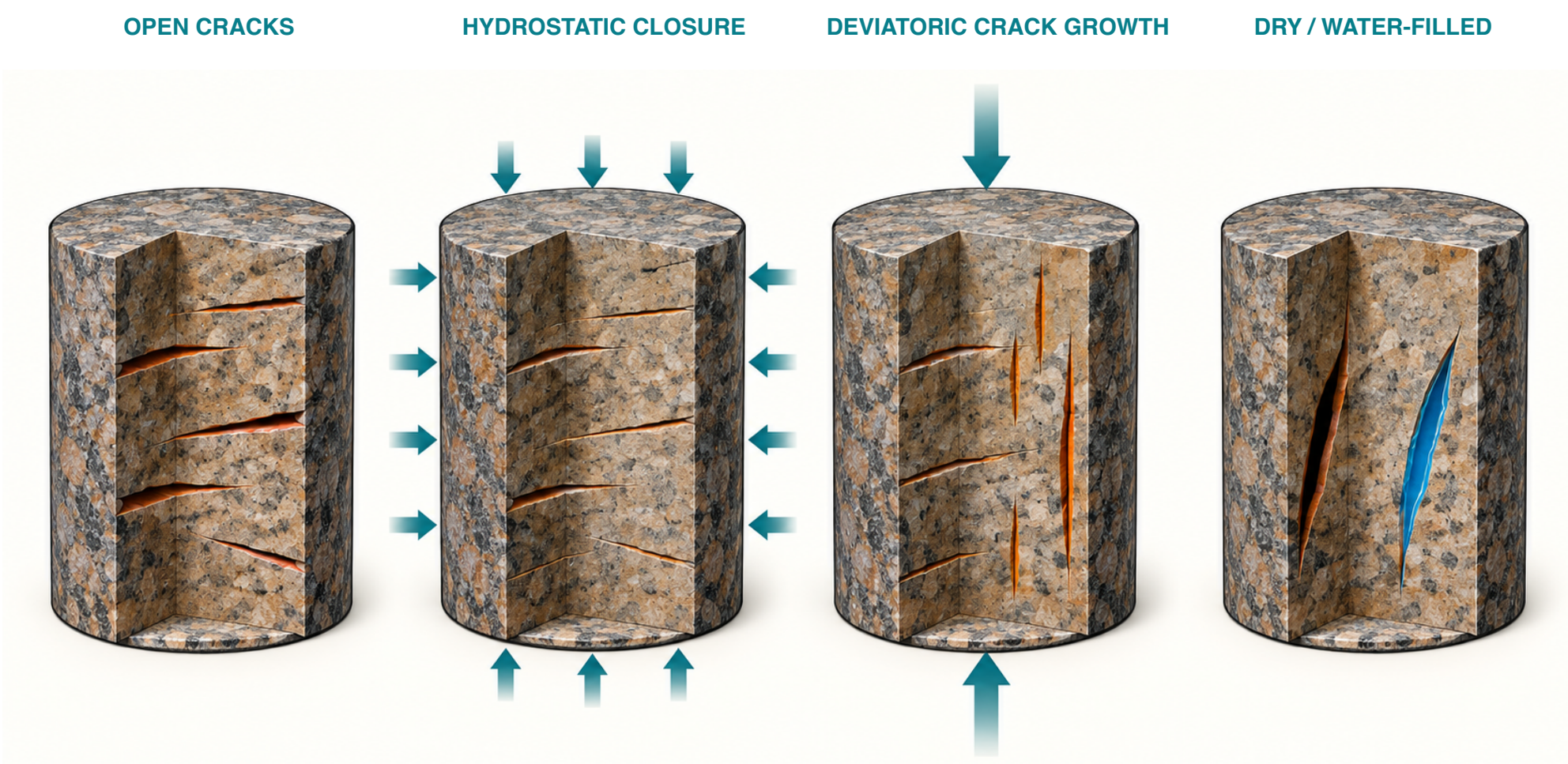
- Moduli calculated from \(V_p\), \(V_s\) and density
- Very small strain and short loading time
- Probes the instantaneous elastic stiffness along the wave path, especially the stiffness of grain contacts and cracks

Why can E_{dyn} exceed E_{static}?

Compliant cracks and contacts may contribute more to quasi-static deformation than to the small-amplitude ultrasonic response.

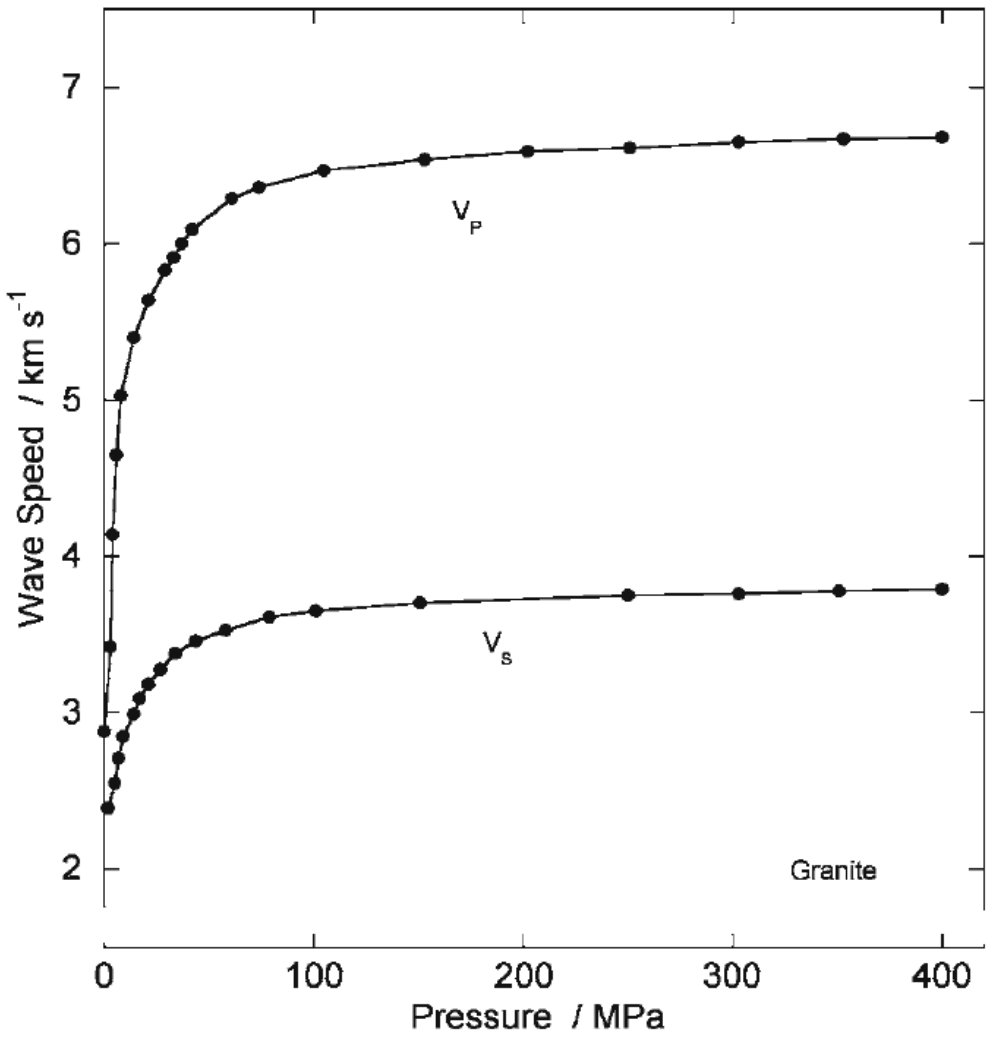
Compare values only when stress state, frequency, strain amplitude and saturation are known.

Cracks make elastic properties sensitive to rock state of stress



Thin cracks contribute little volume but large compliance.

Hydrostatic pressure closes compliant cracks



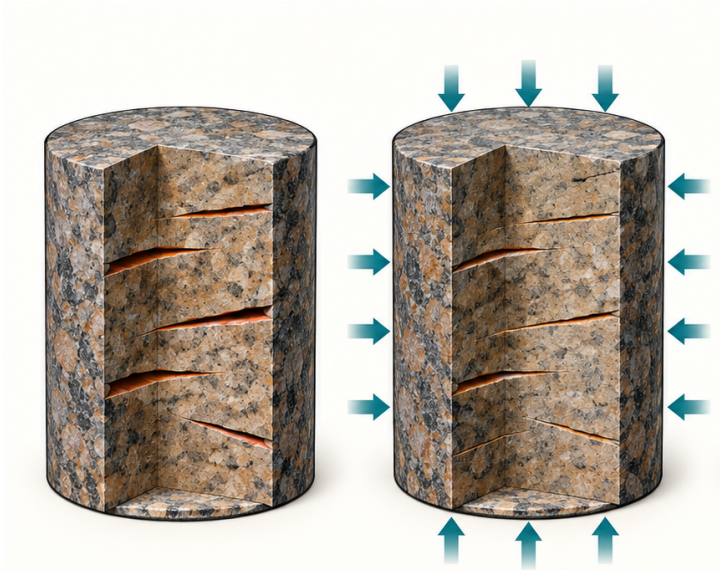
The nonlinear curve records a distribution of crack compliances.

Low pressure

Rapid velocity increase as the most compliant cracks close

High pressure

Smaller additional increase as the response approaches the mineral framework



Prediction check: unloading after crack closure

A dry granite is hydrostatically loaded from 5 to 200 MPa and then unloaded to 5 MPa.

Which velocity path do you predict?

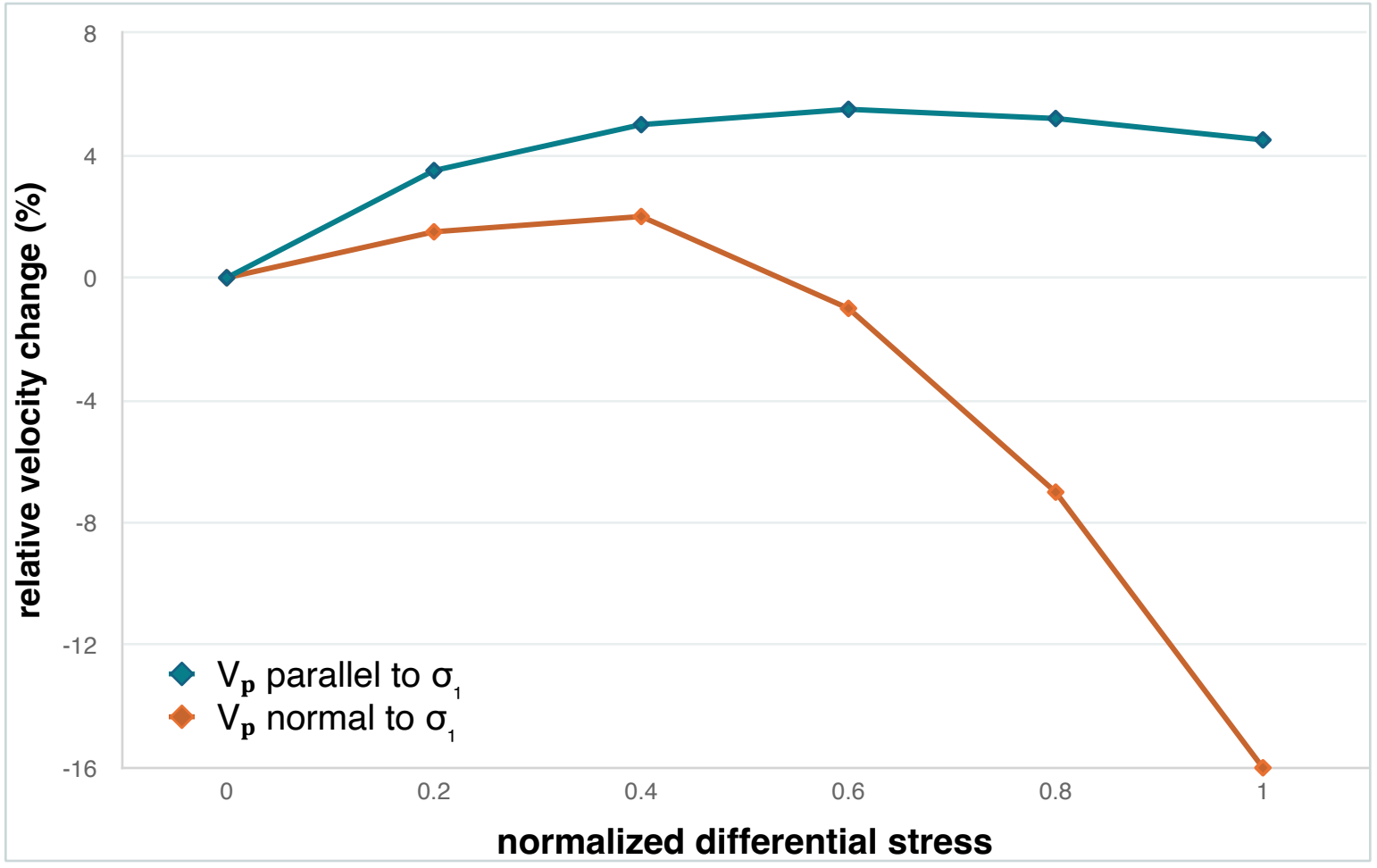
A exactly retraces the loading curve

B returns to a higher velocity because some cracks remain closed or altered

C must return to a lower velocity

D cannot be discussed without specifying stress history and damage

Differential stress creates directional velocity changes



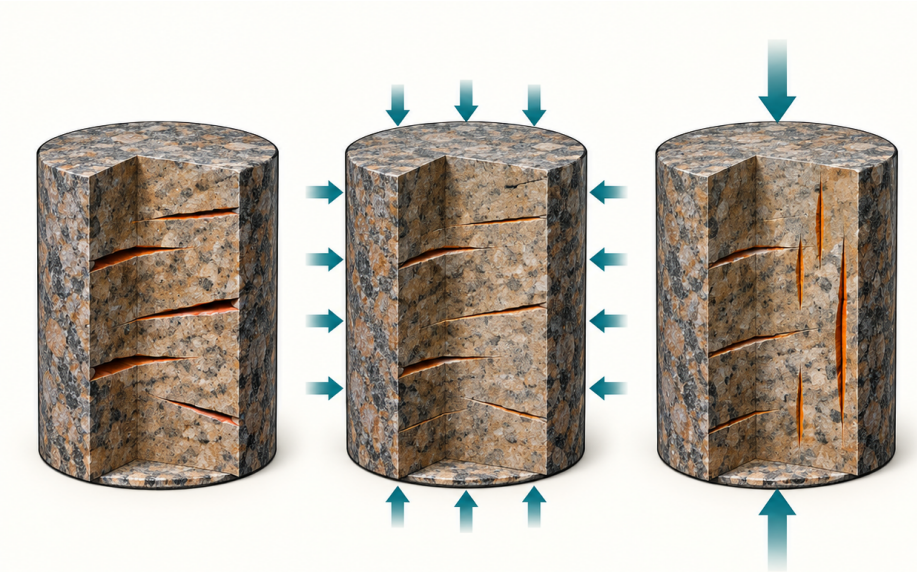
Illustrative trend

EARLY LOADING

Cracks normal to compression close. Both paths may speed up, especially the axial path.

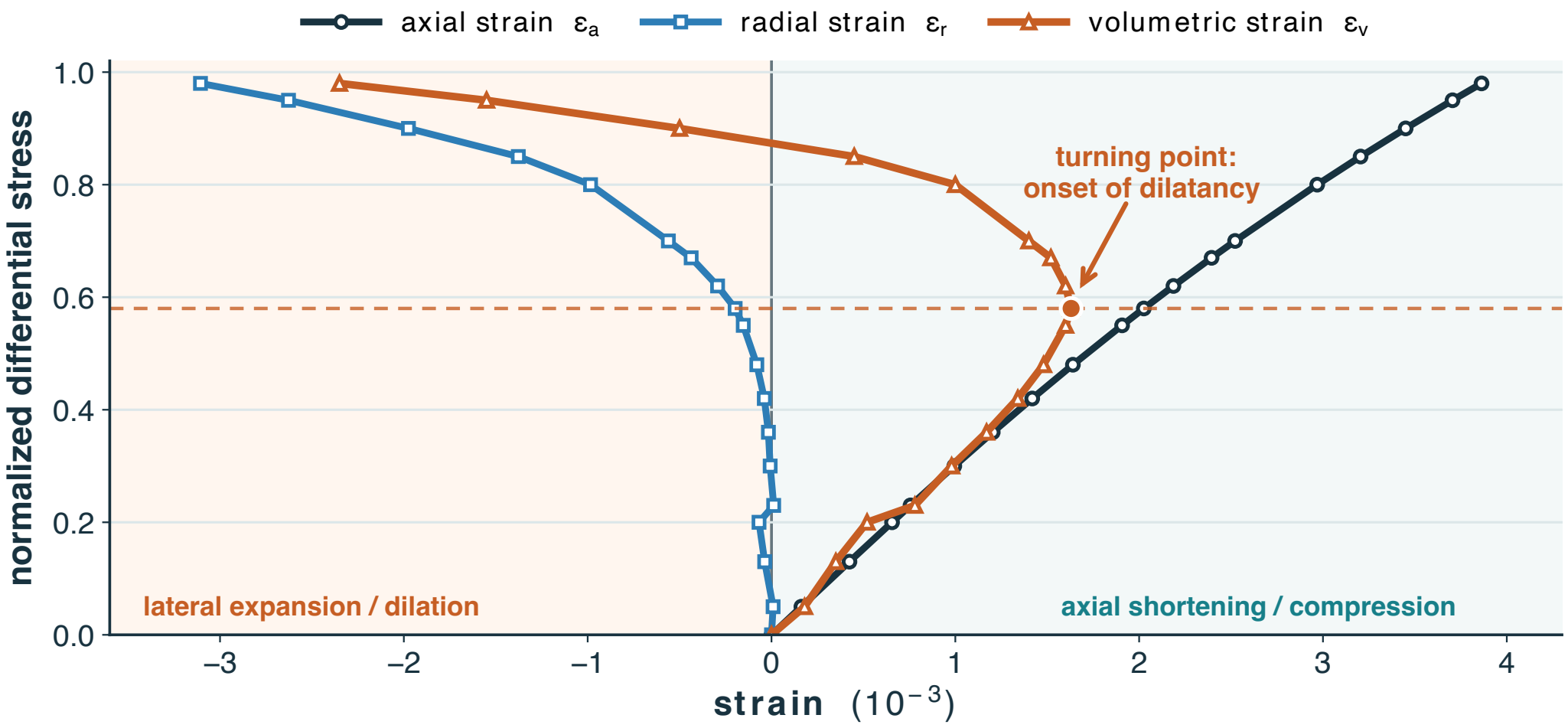
APPROACHING FAILURE

Axial microcracks open. A path normal to σ_1 crosses them and slows markedly.



Velocity changes track dilatancy and crack orientation

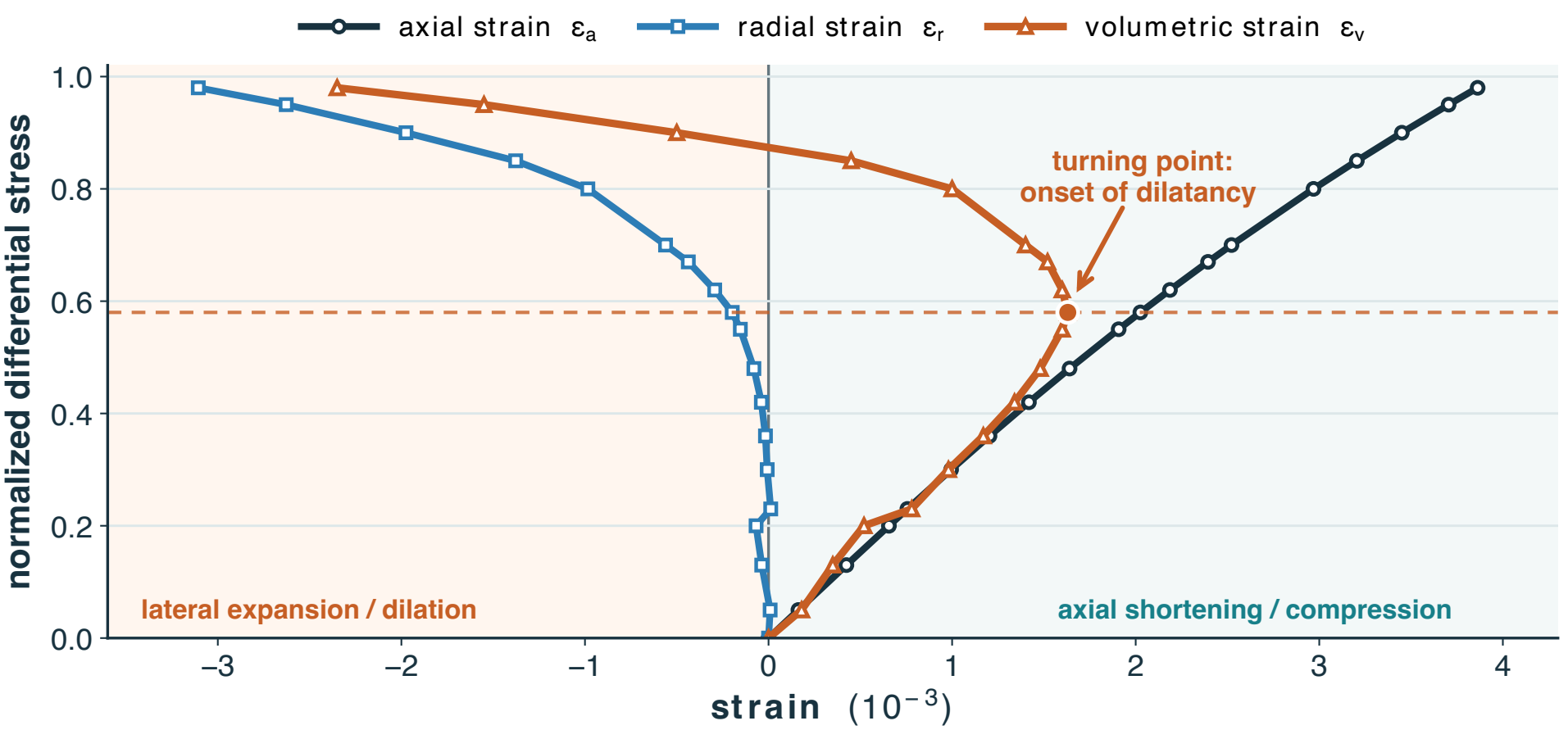
Analyse and comment this plot



Axial and radial strain explain the volumetric response

Velocity changes track dilatancy and crack orientation

Analyse and comment this plot

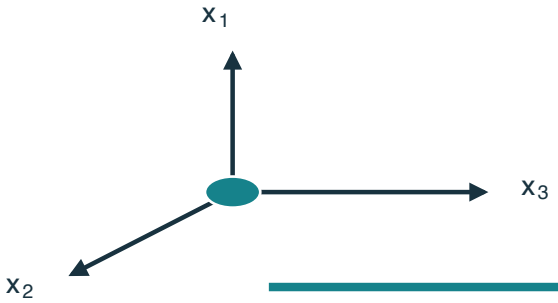




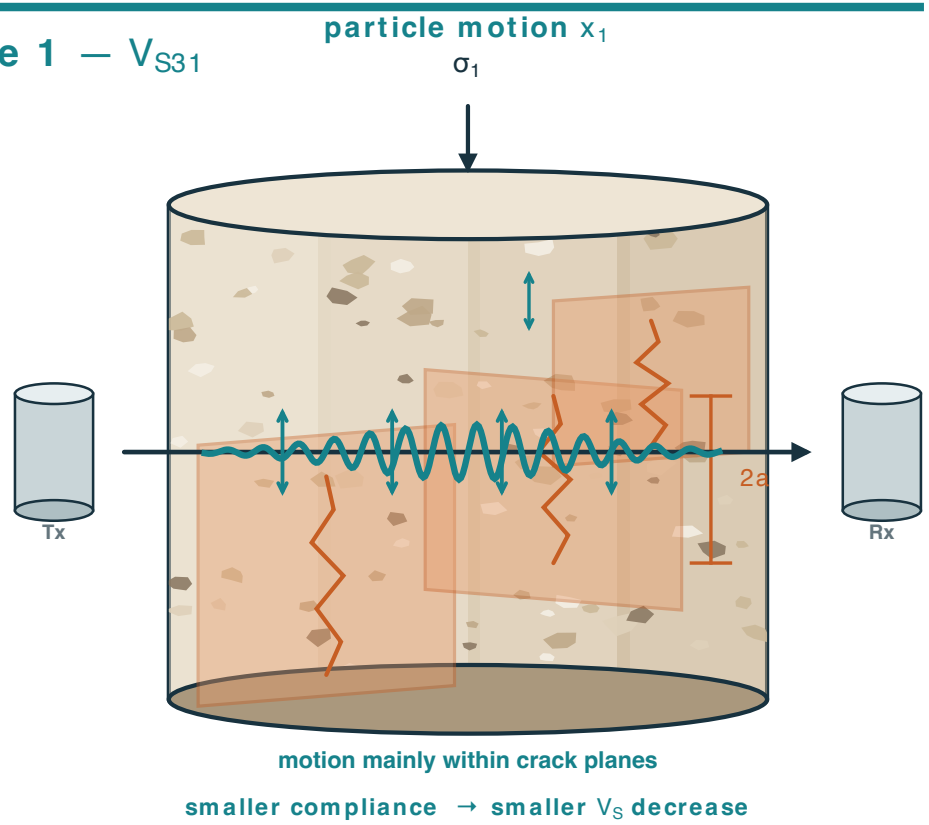
Axial and radial strain explain the volumetric response

Velocity changes track dilatancy and crack orientation

Same propagation path, two S-wave polarizations



Mode 1 — V_{S31}



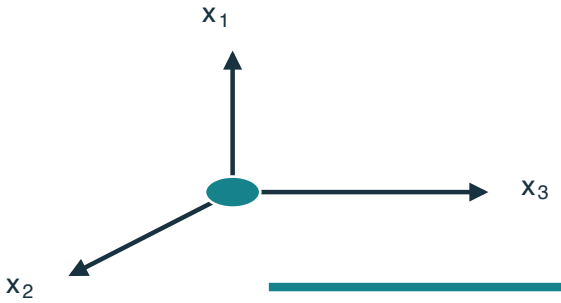
motion mainly within crack planes

smaller compliance → smaller V_S decrease

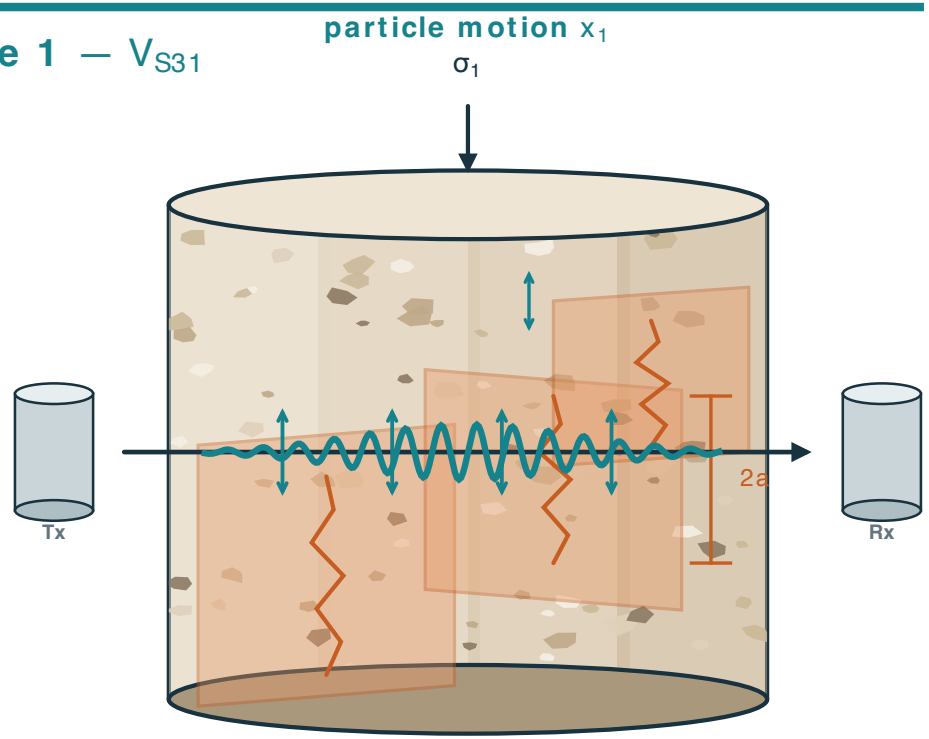


Velocity changes track dilatancy and crack orientation

Same propagation path, two S-wave polarizations



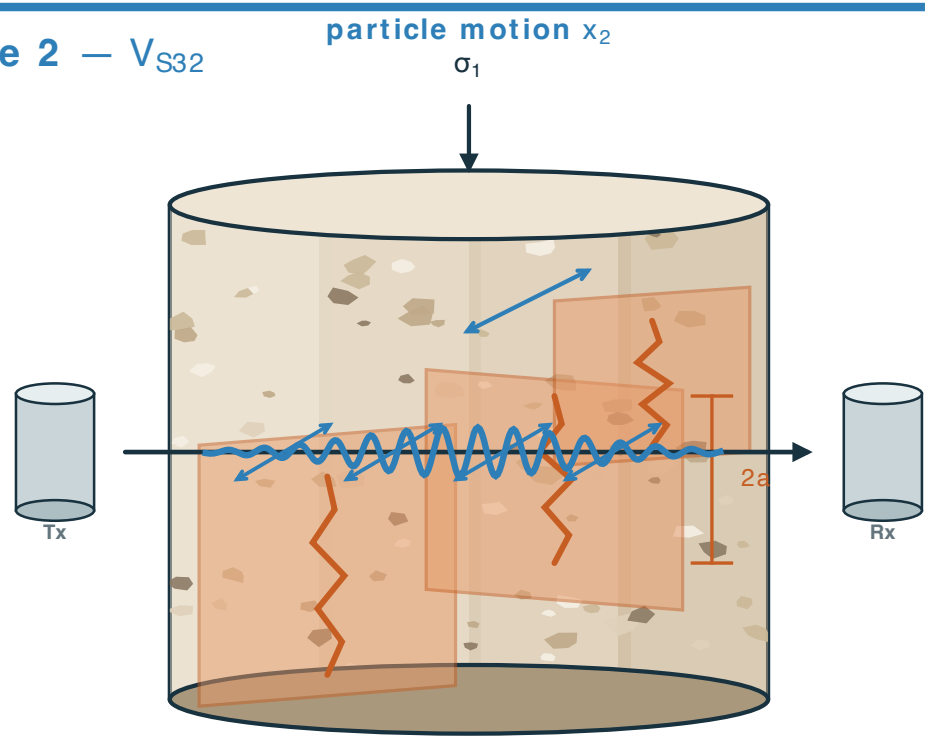
Mode 1 — V_{S31}



motion mainly within crack planes

smaller compliance → smaller V_S decrease

Mode 2 — V_{S32}



motion normal to preferred crack planes

larger compliance → larger V_S decrease

Velocity changes track dilatancy and crack orientation

Same propagation path, two S-wave polarizations

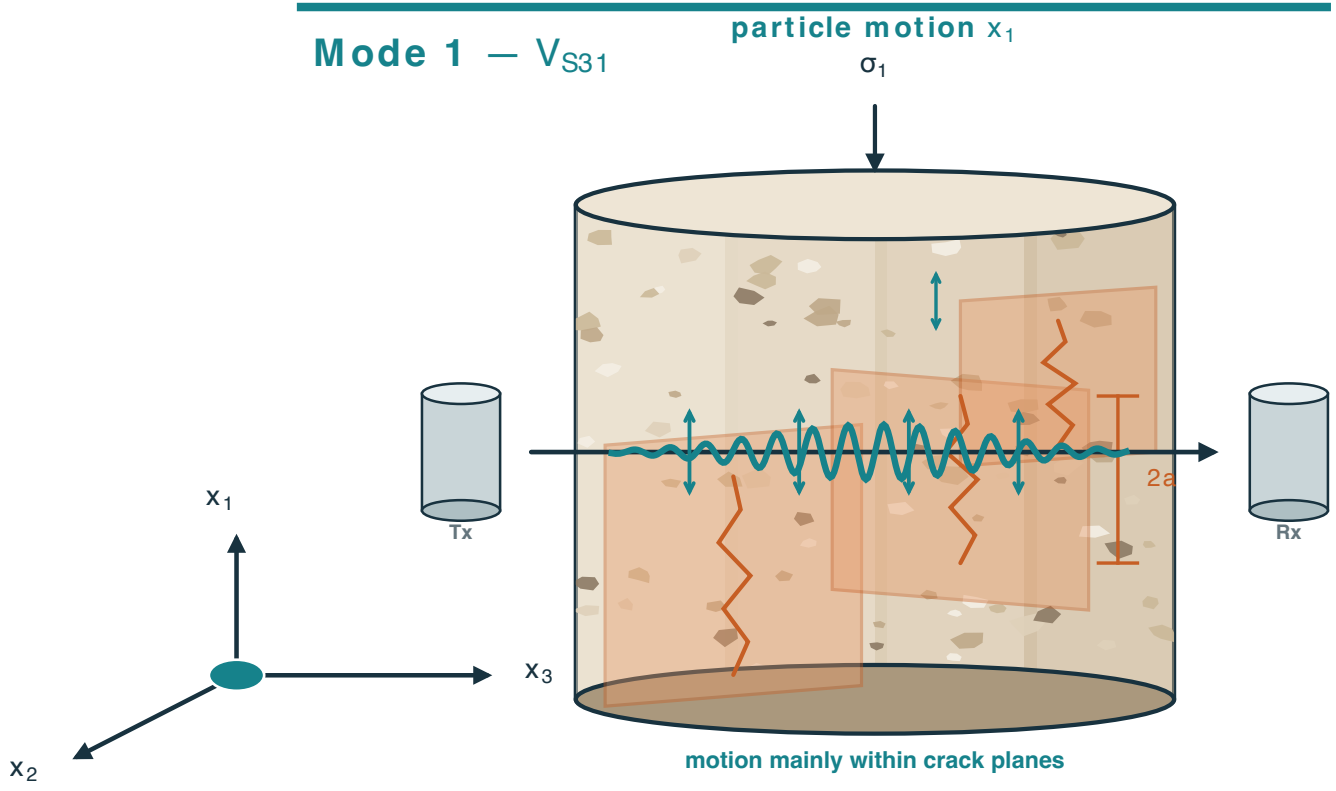
Directional stiffness: $V_{S31} = \sqrt{G_{31}/\rho}$ **and** $V_{S32} = \sqrt{G_{32}/\rho}$

If the preferred crack planes are x_1–x_3, then $G_{32} < G_{31}$ and $V_{S32} < V_{S31}$. The splitting ΔV_S reveals directional crack compliance.

$2a \ll \lambda$

effective-medium response

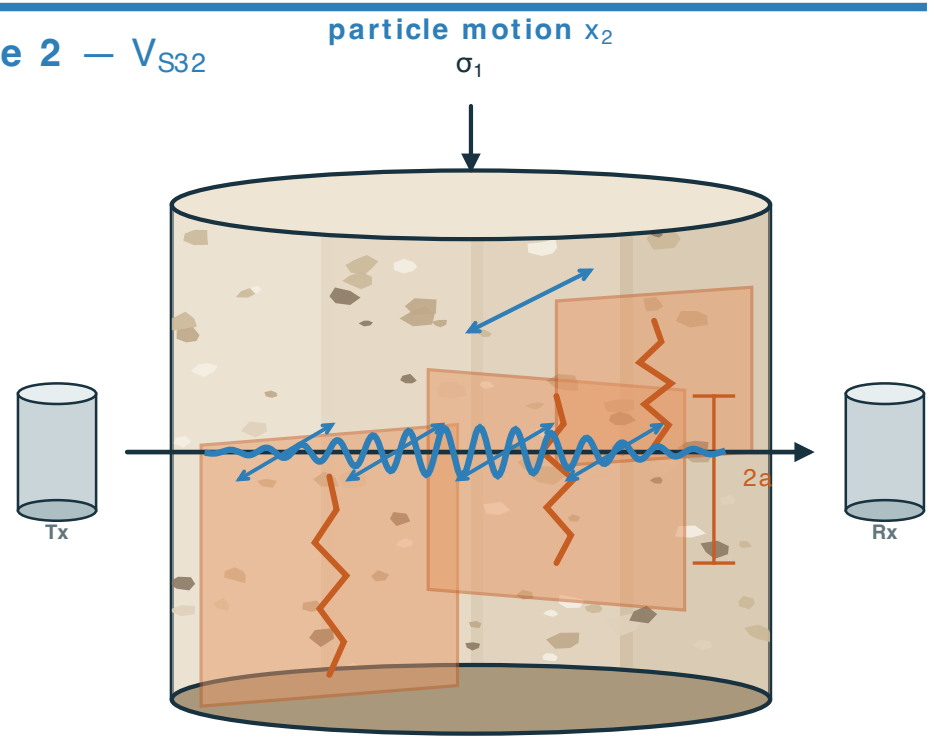
Mode 1 — V_{S31}



motion mainly within crack planes

smaller compliance → smaller V_S decrease

Mode 2 — V_{S32}



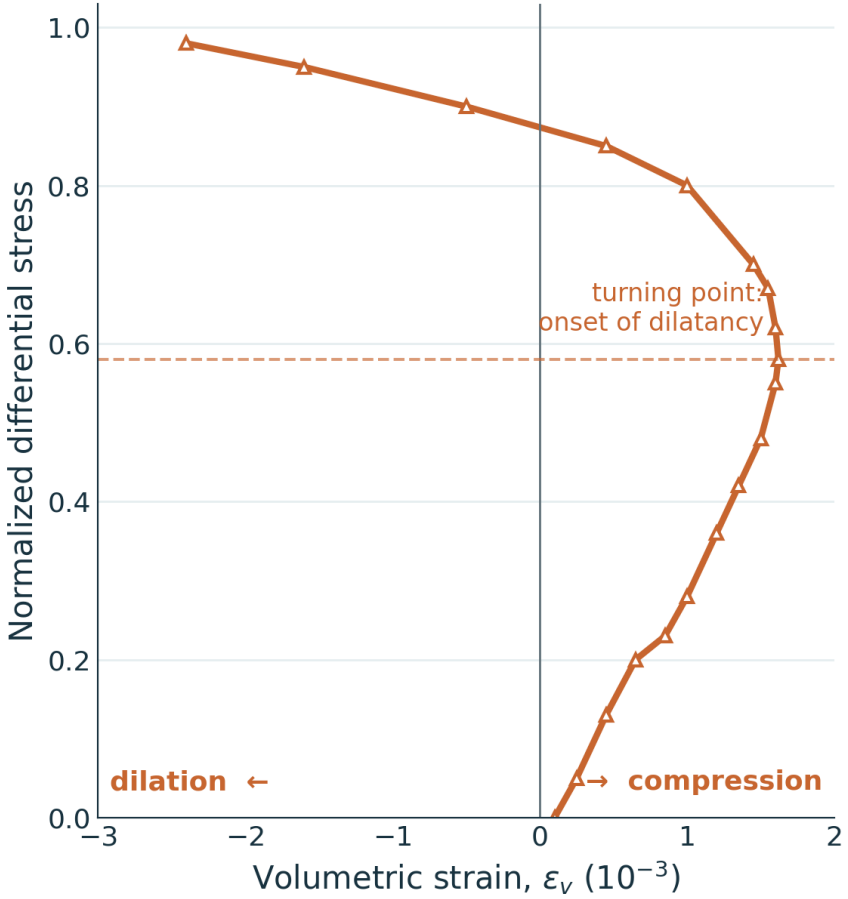
motion normal to preferred crack planes

larger compliance → larger V_S decrease

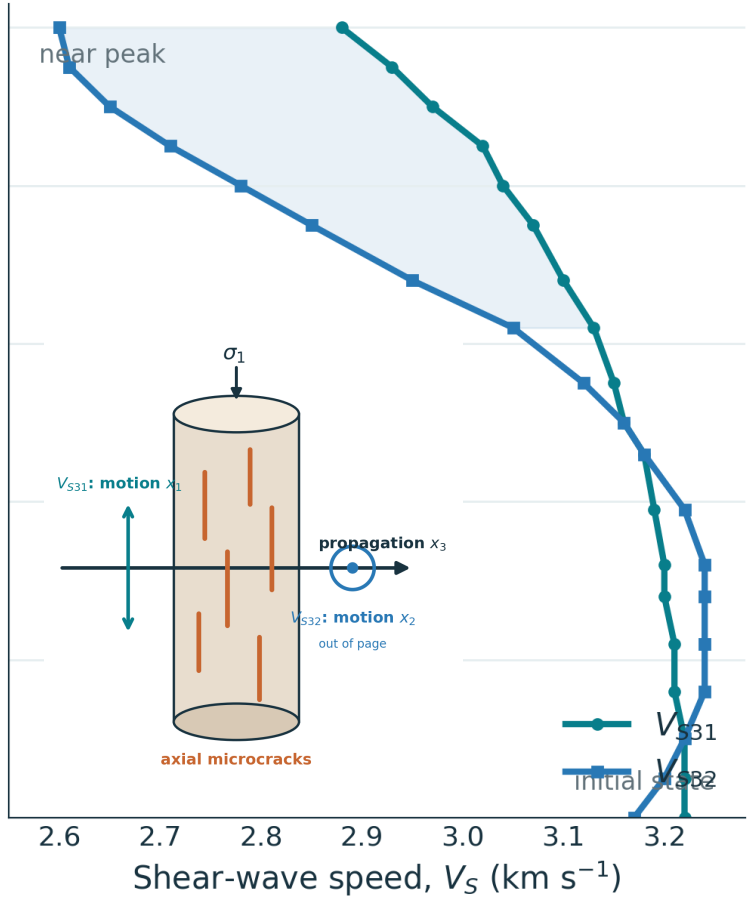
Velocity changes track dilatancy and crack orientation

Dilatancy and shear-wave anisotropy during loading

Volume first compacts, then dilates



The two S-wave modes diverge



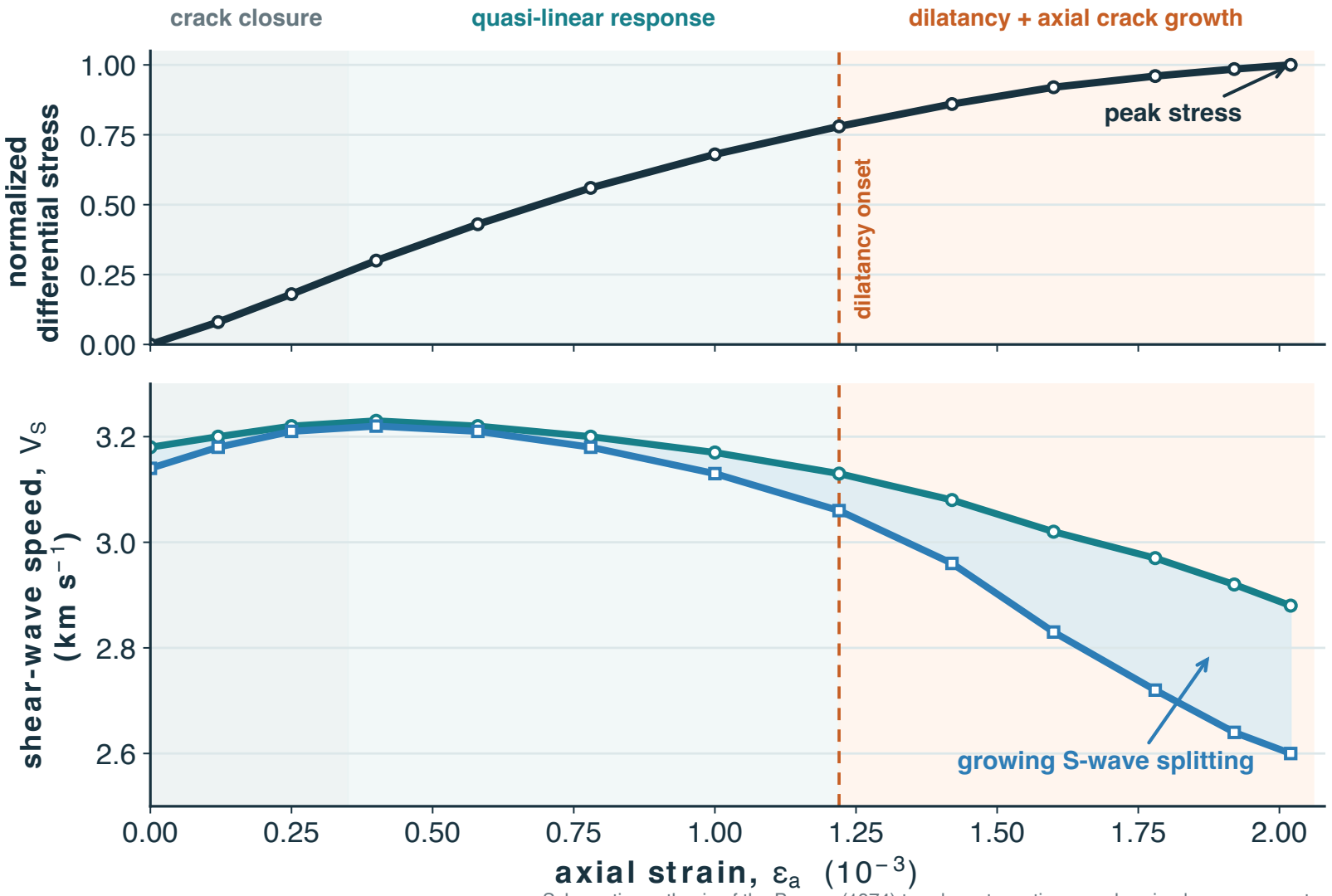
Approximate digitization of Paterson & Wong Fig. 31 (Bonner, 1974)

Read the graph in three passes

1. Locate peak stress.
2. Identify the onset of dilatancy.
3. Compare the two S-wave paths.

Which direction becomes most compliant, and what crack geometry would cause it?

Velocity changes track dilatancy and crack orientation



Schematic synthesis of the Bonner (1974) trends; not raw time-synchronized measurements.

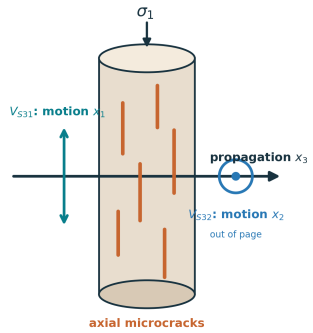
Read the graph in three passes

1. Locate peak stress.
2. Identify the onset of dilatancy.
3. Compare the two S-wave paths.

V_{S31}

V_{S32}

Which direction becomes most compliant, and what crack geometry would cause it?



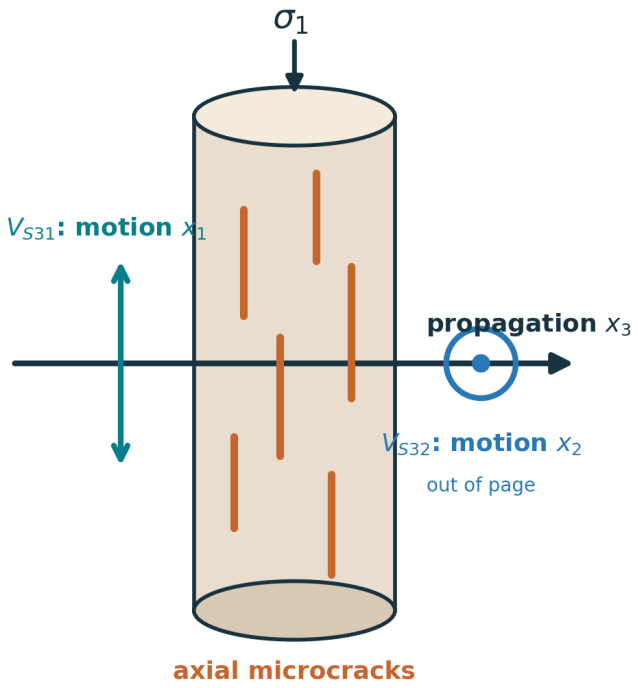
Anisotropy turns propagation direction into information

A single velocity is a path average.

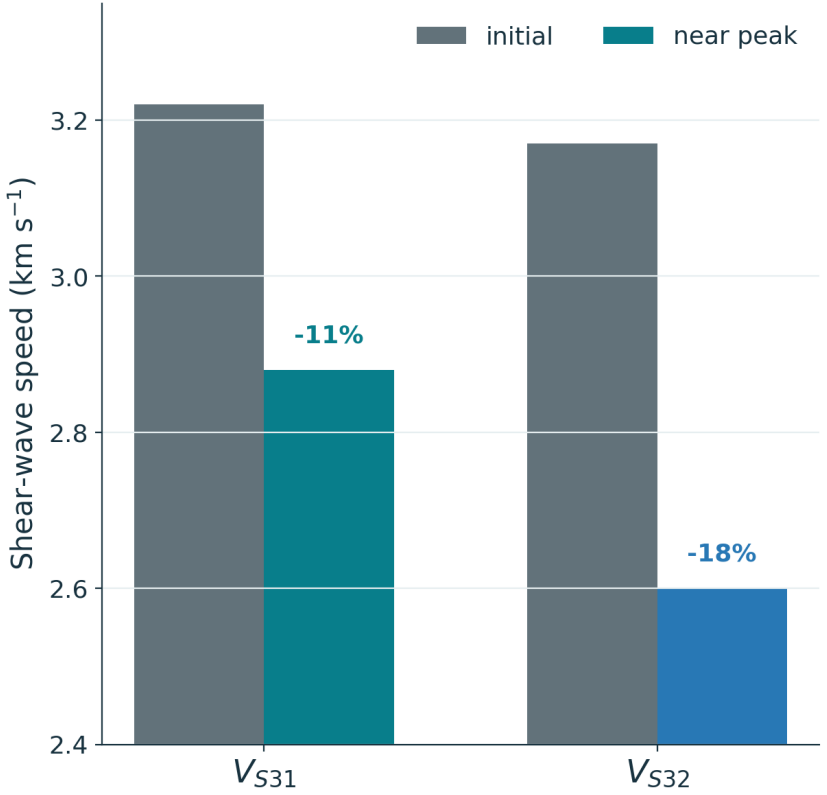
Several paths reveal direction-dependent compliance.

Velocity anisotropy: orientation is encoded in the wave mode

Wave geometry



Measured endpoint change



The larger fall in V_{S32} indicates a larger loss of the corresponding shear stiffness.

Velocity anisotropy may reflect crack orientation, mineral fabric, stress or a combination of them.

Fluid in cracks changes P and S waves differently

DRY CRACKS

- V_p decreases strongly
- V_s also decreases
- V_p/V_s tends to decrease

WATER-FILLED CRACKS

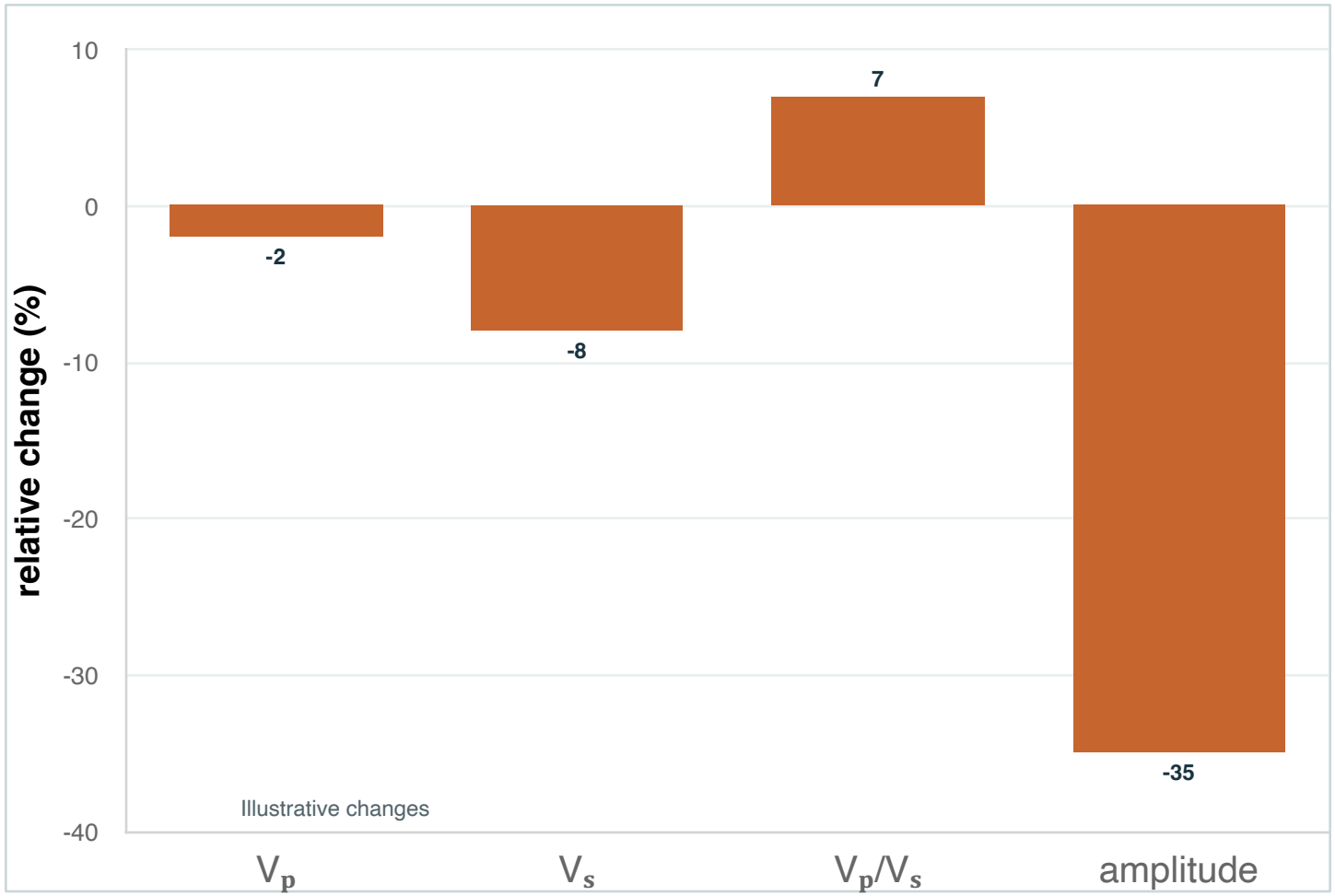
- Fluid resists compression
- V_p recovers strongly
- V_s changes much less
- V_p/V_s tends to increase

V_p/V_s combines crack and fluid sensitivity

The ratio is diagnostic only when mineralogy, stress and crack geometry are constrained.

Paired inference: crack growth or fluid change?

Between two measurements:



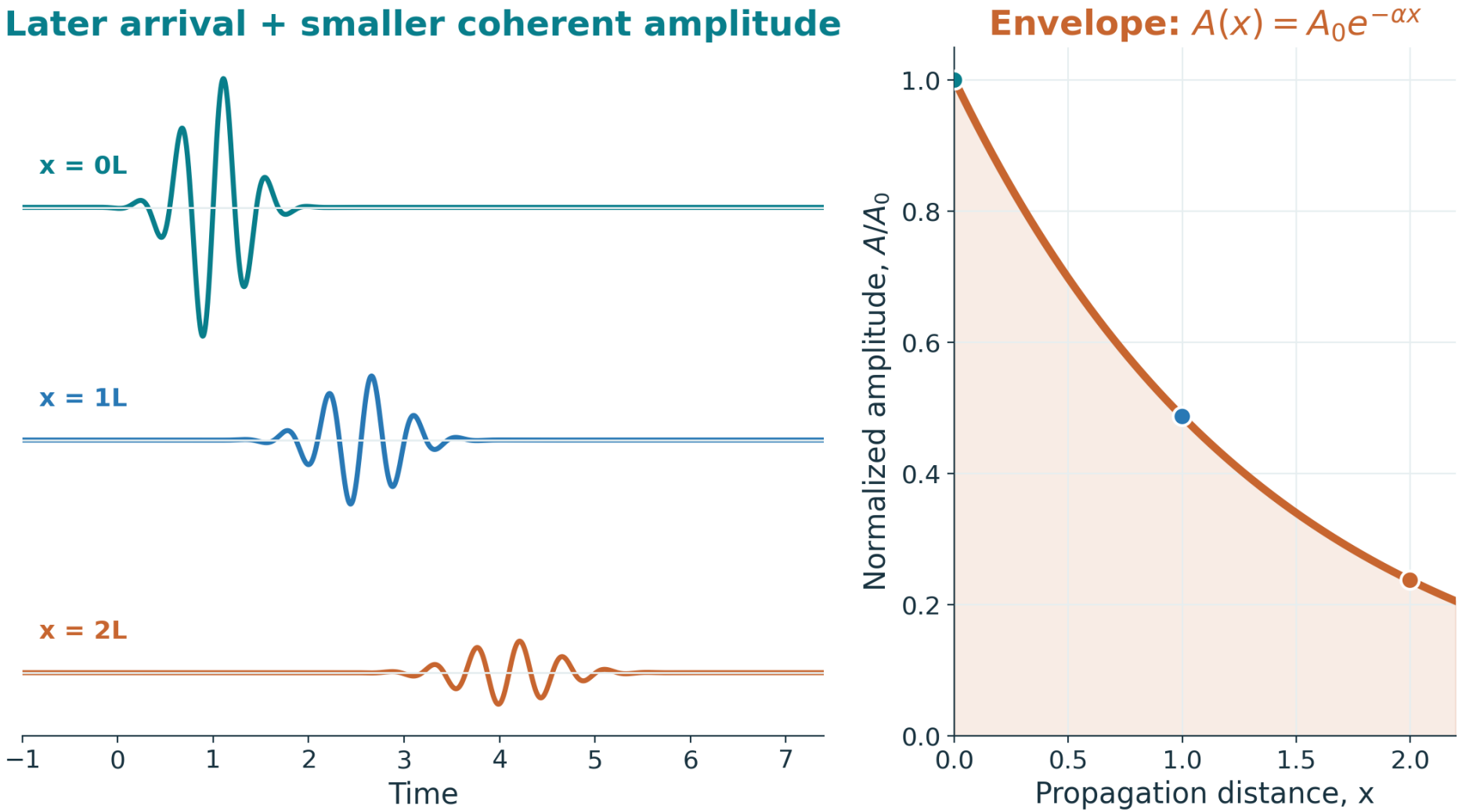
Build two competing explanations

A. more water in existing cracks

B. growth of preferentially oriented cracks

- **Which observation favours each explanation?**
- **What measurement would discriminate them?**

Attenuation measures energy loss from the coherent wave



Intrinsic absorption and scattering can both remove energy from the coherent arrival.

Attenuation measures energy loss from the coherent wave

$$A(x) = A_0 e^{-\alpha x}$$

α describes amplitude decay with distance.

$$Q^{-1} \approx \frac{2\alpha V}{\omega}$$

Q^{-1} is a dimensionless loss measure under a specified convention.

Mechanisms that can increase attenuation

- scattering when heterogeneity approaches λ
- frictional or crack-tip dissipation
- viscous fluid motion in pores and cracks

Attenuation can be more sensitive than velocity, but it is also more vulnerable to coupling and instrument changes.

Travel time, amplitude and coda answer different questions

FIRST ARRIVAL

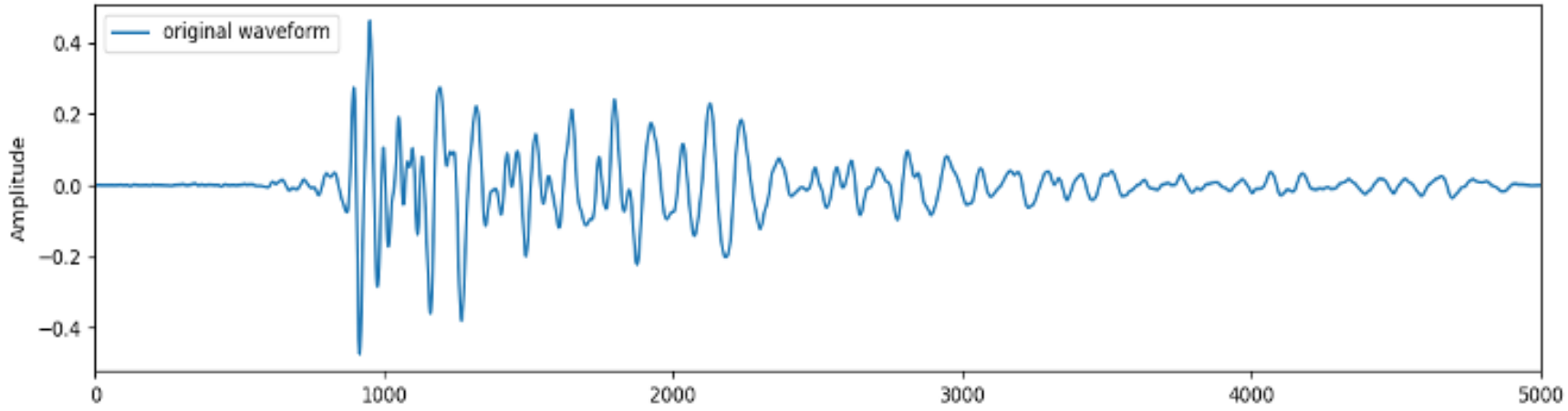
effective stiffness along the fastest coherent path

AMPLITUDE

energy remaining in the selected phase and frequency band

CODA / SPECTRUM

scattering, reverberation and frequency-dependent losses



A complete interpretation keeps the observables separate before combining them.

Later arrival + lower amplitude does not uniquely identify one mechanism.

Amplitude changes require stricter experimental control

COUPLING	contact pressure and couplant thickness
TRANSDUCER	bandwidth, alignment and temperature
ELECTRONICS	gain, filtering and digitizer settings
GEOMETRY	path length, interfaces and specimen deformation

Use attenuation to complement velocity, not to replace it.

Integrated interpretation of elastic-wave observations

Observation	Plausible rock change	Check before concluding
V_p and V_s increase	closure of compliant cracks	pressure path and saturation
transverse V_p decreases	axial crack growth	directional measurements
V_p/V_s increases	fluid-filled cracks	mineralogy and stress
amplitude falls	attenuation or scattering	coupling and gain
coda lengthens	more scattering interfaces	frequency content

These are hypotheses to test, not one-to-one translations.

From waveform to rock-state hypothesis

MEASURE		CALCULATE		COMPARE		INFER
travel time amplitude spectrum	→	V_p, V_s V_p/V_s K, G, E, Poisson ratio	→	direction stress path saturation	→	crack closure crack growth fluid effects

Three takeaways

- V_p and V_s together recover dynamic moduli.
- Directional changes expose crack anisotropy.
- Amplitude strengthens an interpretation only when the measurement chain is controlled.

From waveform to rock-state hypothesis

MEASURE

travel time

amplitude

spectrum

–



→

INFER

crack closure

crack growth

fluid effects

Three takeaways

- V_P
- Di
- Ar

hain is controlled.